NS SAR ADC Project

Revision 1.1 - Project Timeline and Tasks

1. Completed Tasks

- Behavioural modeling of the 8-bit SAR Core.
- Behavioral modeling of Noise Shaping (NS).
- Transistor-level implementation of the analog blocks for the SAR core (Unit Caps, Comparator, Switches).
- Top-level testbench creation

2. Pending Tasks

- Complete a tapeout-ready layout of the SAR core.
- Select the target application.
- Select the NS architecture based on the chosen application (filter orders and active/passive implementation).
- Explore the feasibility of a variable precision (power and accuracy) ADC implementation with NS (8/10/12-bit).
- Familiarize with NS and explore the possibility of an innovation.
- Behavioral modeling of the finalized architecture.
- Transistor-level implementation of the NS block.
- Layout of top-level blocks.
- Decide and implement the DSP for NS (on silicon or firmware).
- SAR control logic implementation (RTL/Synthesis).
- Post-layout verification (DRC, LVS, PEX) of the SAR core.
- System-level PVT verification.

3. Expected Timeline Projection

3.1. Available Time

- **May:** 2 weeks
- **June:** 4 weeks
- **July:** 2 weeks

3.2. Timeline Breakdown

- **May (2 weeks):** Complete tapeout-ready layout of the SAR core, perform SAR logic implementation, run post-layout verification (DRC/LVS/PEX), select target application, and select architecture.
- **June (4 weeks):** Explore innovation in the selected architecture, perform behavioral modeling, transistor modeling, and layout.
- **July (2 weeks):** Finalize NS layout, complete top-level layout, run top-level testbench and system-level PVT verification, and finalize DSP on silicon decision and implementation.